

AI Content Moderation Platform

Intern Assignment – Full Stack

1. Introduction

Design and implement a full-stack AI-powered Content Moderation Platform that allows users to submit images for automated policy compliance screening. The system supports a structured appeal process for disputed verdicts, gives administrators meaningful oversight tools, and allows fine-grained control over how moderation policies are enforced across different violation categories.

2. Objectives

- Demonstrate the ability to build a complete, multi-role web application from scratch.
- Integrate an AI model meaningfully into a core workflow
- Show proficiency in data modeling across interconnected entities: users, submissions, verdicts, appeals, and policies.
- Apply best practices in project structure, API design, and code organization.
- Deliver a functional frontend that accurately reflects real system state at all times.

3. Moderation Categories

Every submitted image is screened against all active moderation categories. For each category, the system produces a classification result, a confidence score, and a short reasoning summary.

Category	Description
Graphic Violence	Depictions of physical harm, gore, or serious injury to humans or animals.
Hate Symbols	Imagery associated with extremist ideologies or designated terrorist organizations.
Self-Harm	Visual content depicting or glorifying acts of self-inflicted injury.
Extremist Propaganda	Content that promotes, recruits for, or glorifies violent extremist movements.
Weapons & Contraband	Imagery depicting illegal weapons, drug manufacturing, or trafficking-related content.
Harassment & Humiliation	Imagery intended to degrade, threaten, or publicly humiliate an identifiable individual.

4. Core Features

4.1 User Submissions

Users can submit one or more images in a single request. Each image is screened independently and receives its own verdict. A submission is considered clean if no active category produces a result that meets or exceeds the configured confidence threshold. Users can view their full submission history and filter results by outcome, category, and date.

4.2 Verdict System

Each image produces a structured verdict containing the following:

- The overall outcome: Approved, Flagged for Review, or Blocked.
- A per-category breakdown listing the classification result, confidence score, and a brief reasoning string.
- A timestamp and a reference to the policy configuration active at the time of the submission.

The overall outcome is determined by the enforcement behavior configured per category by an administrator. If a category is set to Auto-Block, any result meeting the confidence threshold immediately marks the submission as Blocked. If set to Flag for Review, the submission is marked as Flagged and enters the admin review queue.

4.3 Appeal Workflow

Users may file an appeal against any submission that received a Flagged or Blocked outcome. An appeal requires a written justification from the user explaining why they believe the verdict is incorrect. Each appeal is linked to a specific submission and enters a pending queue visible only to administrators.

Administrators review each appeal and either accept or reject it, with the option to attach a written response. Upon acceptance, the submission verdict is overridden to Approved. Users can track the current status of any appeal (Pending, Accepted, or Rejected) from their submission history.

4.4 Policy Configuration

Administrators control the behavior of each moderation category independently. For every category, an admin may:

- Enable or disable the category entirely. Disabled categories are skipped during screening.
- Set a confidence threshold, expressed as a percentage, below which a detection is treated as inconclusive and does not affect the verdict.
- Set the enforcement behavior to either Auto-Block or Flag for Review.

All policy changes apply to submissions made after the change and do not retroactively alter existing verdicts.

4.5 Admin Analytics Dashboard

Administrators have access to a platform-wide analytics view that provides a clear picture of system activity. This includes:

- Total submission volume over time.
- Verdict distribution broken down by outcome and by category.
- Appeal volume, resolution rate, and outcome breakdown (accepted vs. rejected).
- A ranked list of users by submission count and by violation count.

5. User Roles

The system has two distinct roles with non-overlapping administrative capabilities.

Role	Capabilities
User	Register, log in, submit images, view personal submission history, file appeals, and track appeal status.
Admin	All user capabilities, plus access to the appeals queue, manual verdict overrides, policy configuration, and the analytics dashboard.

6. Deliverables

- A fully functional web application with a connected frontend and backend.
- A REST API that serves as the sole interface between the frontend and the database.
- A MongoDB database with a well-structured, documented schema.
- A Dockerized setup that runs completely with a single docker-compose up command.
- A README covering setup instructions, required environment variables, and a brief explanation of key architecture decisions.